

Each film coated tablet contains:

Memantine hydrochloride 10 mg eq. to Memantine 8.31 mg

✧ PRODUCT DESCRIPTION:

White, centrally tapered, oblong, biconvex, film coated tablet with engraved MMT on one side and scored on the other

✧ MECHANISM OF ACTION:

Pharmacology

There is increasing evidence that malfunctioning of glutamatergic neurotransmission, in particular at NMDA-receptors, contributes to both expression of symptoms and disease progression in neurodegenerative dementia.

Memantine is a voltage-dependent, moderate-affinity uncompetitive NMDA-receptor antagonist. It modulates the effects of pathologically elevated tonic levels of glutamate that may lead to neuronal dysfunction.

Pharmacokinetics

Absorption

Memantine has an absolute bioavailability of approximately 100%. t_{max} is between 3 and 8 hours. There is no indication that food influences the absorption of Memantine.

Distribution

Daily doses of 20 mg lead to steady-state plasma concentrations of Memantine ranging from 70-150 ng/mL (0.5-1 μ mol) with large interindividual variations. When daily doses of 5-30 mg were administered, a mean cerebrospinal fluid (CSF)/serum ratio of 0.52 was calculated. The volume of distribution is around 10 L/Kg. About 45% of Memantine is bound to plasma-proteins.

Biotransformation

In man, about 80% of the circulating Memantine-related material is present as the parent compound. Main human metabolites are N-3,5-dimethyl-glutantan, the isomeric mixture of 4-and 6-hydroxy-memantine, and 1-nitroso-3,5-dimethyl-adamantane. None of these metabolites exhibit NMDA-antagonistic activity. No cytochrome P 450 catalyzed metabolism has been detected *in vitro*.

Elimination

Memantine is eliminated in a monoexponential manner with a terminal $t_{1/2}$ of 60-100 hours. In volunteers with normal kidney function, total clearance (Cl_{tot}) amounts to 170 mL/min/1.73 m² and part of total renal clearance is achieved by tubular secretion. Renal handling also involves tubular reabsorption, probably mediated by cation transport proteins. The renal elimination rate of Memantine under alkaline urine conditions may be reduced by a factor of 7-9. Alkalization of urine may result from drastic changes in diet, e.g. from a carnivore to a vegetarian diet, or from the massive ingestion of alkalinizing gastric buffers.

Linearity

Studies in volunteers have demonstrated linear pharmacokinetics in the dose range of 10-40 mg.

Pharmacokinetic/pharmacodynamic relationship

At a dose of Memantine of 20 mg per day the CSF levels match the k_i -value (k_i = inhibition constant) of Memantine, which is 0.5 μ mol in human frontal cortex.

✧ INDICATION:

Indicated for the treatment of patients with moderate to severe Alzheimer's disease.

✧ DOSAGE AND ADMINISTRATION:

Oral

Treatment should be initiated and supervised by a physician experienced in the diagnosis and treatment of Alzheimer's dementia. Therapy should only be started if a caregiver is available who will regularly monitor the intake of the medicinal product by the patient. Diagnosis should be made according to current guidelines. The tolerance and dosing of Memantine should be reassessed on a regular basis, preferably within three months after start of treatment. Thereafter, the clinical benefit of Memantine and the patient's tolerance of treatment should be reassessed on a regular basis according to current clinical guidelines. Maintenance treatment can be continued for as long as a therapeutic benefit is favorable and the patient tolerates treatment with Memantine. Discontinuation of Memantine should be considered when evidence of a therapeutic effect is no longer present or if the patient does not tolerate treatment.

Posology

Memantine should be administered once a day and should be taken at the same time every day.

Adults

Dose titration

The maximum daily dose is 20 mg per day. In order to reduce the risk of undesirable effects the maintenance dose is achieved by upward titration of 5 mg per week over the first 3 weeks as follows:

Week 1 (day 1-7): The patient should take half a 10 mg film coated tablet (5 mg) per day for 7 days.

Week 2 (day 8-14): The patient should take one 10 mg film coated tablet (10 mg) per day for 7 days.

Week 3 (day 15-21): The patient should take one and a half 10 mg film coated tablet (15 mg) per day for 7 days.

From Week 4 on: The patient should take two 10 mg film coated tablets (20 mg) per day.

Maintenance dose

The recommended maintenance dose is 20 mg per day.

Elderly

On the basis of the clinical studies, the recommended dose for patients over the age of 65 years is 20 mg per day (two 10 mg tablets once a day) as described above.

Pediatric population

Memantine is not recommended for use in children below 18 years due to a lack of data on safety and efficacy.

Renal impairment

In patients with mildly impaired renal function (creatinine clearance 50-80 mL/min) no dose adjustment is required. In patients with moderate renal impairment (creatinine clearance 30-49 mL/min) daily dose should be 10 mg per day. If tolerated well after at least 7 days of treatment, the dose could be increased up to 20 mg/day according to standard titration scheme. In patients with severe renal impairment (creatinine clearance 5-29 mL/min) daily dose should be 10 mg per day.

Hepatic impairment

In patients with mild or moderate hepatic impaired function (Child-Pugh A and Child-Pugh B) no dose adjustment is needed. No data on the use of Memantine in patients with severe hepatic impairment are available. Administration of Memantine is not recommended in patients with severe hepatic impairment.

Method of administration

The film coated tablets can be taken with or without food.

✧ PRECAUTION:

Caution is recommended in patients with epilepsy, former history of convulsions or patients with predisposing factors for epilepsy. Concomitant use of N-methyl-D-aspartate (NMDA)-antagonists such as Amantadine, Ketamine or Dextromethorphan should be avoided. These compounds act at the same receptor system as Memantine, and therefore adverse reactions (mainly central nervous system (CNS)-related) may be more frequent or more pronounced. Some factors that may raise urine pH may necessitate careful monitoring of the patient. These factors include drastic changes in diet, e.g. from a carnivore to a vegetarian diet, or a massive ingestion of alkalinizing gastric buffers. Also, urine pH may be elevated by states of renal tubular acidosis (RTA) or severe infections of the urinary tract with *Proteus* bacteria.

Product	Code No.	Dimension	Packaging Type	Thickness
MEMXA 10	IMMY 0051	W 16.5 x L 28.5 cm.	Wood Free Paper (non-embossed)	60 g (0.08 mm)
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